

Optimal Loading for Maximal Power Output during Lower-Body Resistance Exercises

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ABSTRACT

CORMIE, P., G. O. MCCAULLEY, N. T. TRIPLETT, and J. M. MCBRIDE. Optimal Loading for Maximal Power Output during Lower-Body Resistance Exercises. *Med. Sci. Sports Exerc.*, Vol. 39, No. 2, pp. 340–349, 2007. **Purpose:** The influence of various loads on power output in the jump squat (JS), squat (S), and power clean (PC) was examined to determine the load that maximizes power output in each lift. **Methods:** Twelve Division I male athletes participated in four testing sessions. The first session involved performing one-repetition maximums (1RM) in the S and PC, followed by three randomized testing sessions involving either the JS, S, or PC. Peak force, velocity, and power were calculated across loads of 0, 12, 27, 42, 56, 71, and 85% of each subject's 1RM in the JS and S and at 10% intervals from 30 to 90% of each subject's 1RM in the PC. **Results:** The optimal load for the JS was 0% of 1RM; absolute peak power was significantly lower from the optimal load at 42, 56, 71, and 85% of 1RM ($P \leq 0.05$), whereas peak power relative to body mass was significantly lower at 27% of 1RM in addition to 42, 56, 71, and 85% of 1RM. Peak power in the S was maximized at 56% of 1RM; however, power was not significantly different across the loading spectrum. The optimal load in the PC occurred at 80% of 1RM. Relative peak power at 80% of 1RM was significantly different from the 30 and 40% of 1RM. **Conclusion:** This investigation indicates that the optimal load for maximal power output occurs at various percentages of 1RM in the JS, S, and PC. **Key Words:** SQUAT, JUMP SQUAT, POWER CLEAN, VELOCITY, FORCE

Kaneko et al. (12) performed one of the first studies investigating the effect of loading during resistance exercise on maximal power output. It was reported that using a resistance equal to 30% of maximal isometric strength in an elbow flexor movement maximized power output. Additionally, training with a load that maximizes power output (the optimal load) was shown to result in the greatest increases in muscle power (12). Therefore, identifying the optimal load in lifts commonly used for power training is critical. Although consistency regarding the identification of the load that maximizes power output exists in investigations involving *in vitro* study of muscle fibers (4,26) and single-joint movements (12,20), investigations of dynamic, lower-body resistance exercises have reported differing results. The load that maximizes power output in the jump squat (JS) has been extensively investigated (1,5,16,25) but has been reported to range from 0% (16) to 60% of one-repetition maximum

(1RM) (1). The discrepancies between these previous investigations may be attributable to variations in the methodological procedures or differences in the type of subjects used for each study. Although far fewer data exist concerning the optimal load in the squat (S) and power clean (PC), previous studies have reported a load of approximately 60% of 1RM for the S (10,11,23) and 70–80% of 1RM for the PC (8,28). Therefore, it was hypothesized that the load that optimizes power output would occur at approximately 0% of 1RM in the JS, 60% 1RM in the S, and 70–80% of 1RM in the PC.

The load–power relationship is theorized to differ between the JS, S, and PC because of the nature of movements involved with the respective lifts. The JS, similar in nature to a bench-press throw, is considered a “ballistic” exercise in that the deceleration phase is much smaller in comparison with the standard S movement, where the bar is not released or thrown (21). Because of this observed phenomenon, the load that maximizes power output (e.g., in an S), would most likely be heavier than the load that maximizes power output in a JS because of differences in the maximal velocity achieved at each respective load (11,21). The PC is a modified weightlifting exercise that is commonly used in resistance training programs. A paradigm constructed from separate investigations allows for speculation that the deceleration phase of the PC would fall somewhere between the JS and S (7). However, because of the high-force and high-velocity nature of the PC, maximal power output may occur at a

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relatively high percentage of 1RM (8). Thus, the load that would maximize power output in the PC would likely occur at a higher percentage of 1RM than the optimal load in the JS and, possibly, the S. The current investigation is the first to examine these three lifts in the context of a single investigation.

The relationship between power and dynamic athletic performance has been well established in that muscular power is regarded as one of the primary factors involved in sports that entail the production of high forces over a short time (14,21). Improvements in maximal power output have been accompanied by increased performance in jumping, sprinting, and agility tests, as well as beneficial changes in rate of force development (17,28). Training at the load that maximizes power output is the most effective stimulus for further improvements in muscle power capabilities (12,17,28). Kaneko et al. (12) reported that individuals training with 30% of maximal isometric force in an elbow flexor movement for a period of 12 wk increased maximal power output by an average of 26%, which was higher than the result of training with 0, 60, or 100% of maximal isometric force. Therefore, it is reasonable to assume that in other exercises such as the JS, S, and PC, the most effective intensity for training would be a load that maximizes power output in each of the respective lifts. This assumption has been supported by investigations of the JS in which training with the load that maximized power output resulted in greater improvements in maximal power output than other intensities (17,27). Because the optimal load varies according to the nature of a movement, it is imperative that the load–power relationship of various lifts be examined to establish effective, exercise-specific training recommendations.

No previous investigations have simultaneously compared the optimal load for maximizing power output in the JS, S, and PC. This is the most unique aspect of the current investigation. In addition, it could be argued that previous methodology has not been adequate in accurately assessing power output in dynamic lower-body resistance exercises. Recent data confirm that measurement of power through the use of only a force plate or a single linear position transducer, which has been commonly used in previous studies, may not be valid (6). Therefore, the purpose of this investigation was to assess the load that would maximize power output in the JS, S, and PC, using appropriate data-collection and analysis procedures.

METHODS

Subjects. Twelve Division I male athletes (football players, sprinters, long jumpers) participated in this study (age: 19.83 ± 1.40 yr; height: 179.10 ± 4.56 cm; weight: 90.08 ± 14.81 kg; % body fat: $11.85 \pm 5.47\%$; S 1RM: 170.38 ± 21.72 kg; PC 1RM: 112.50 ± 13.15 kg; S 1RM–body weight ratio: 1.91 ± 0.22). The participants were notified about the potential risks involved and gave their written informed consent, approved by the institutional review board at Appalachian State University.

Study design. Subjects visited the laboratory on four separate occasions. The first session involved 1RM testing in the S and PC. They were then familiarized with the testing procedures for the subsequent sessions involving the JS, S, and PC. The following three sessions were randomized and involved performing the respective lifts with loads ranging from 0, 12, 27, 42, 56, 71, and 85% of each subject's S 1RM for the sessions involving the JS and S, and at 10% intervals from 30 to 90% of each subject's PC 1RM for the session involving the PC. These loading conditions were used because they represent 30, 40, 50, 60, 70, 80, and 90% of maximal dynamic strength ($MDS = 1RM + [\text{body mass} - \text{shank mass}]$) in the JS, S, and PC. Because the mass of the body must be incorporated with the external load when determining the system mass during lower-body movements, MDS represents the total load being accelerated. Therefore, MDS incorporates both the external load and an individual's body mass, whereas the 1RM represents the external load only (i.e., 0% of 1RM [body mass only] is equivalent to 30% of MDS [body mass only]). The external load differs between the PC and both the JS and S, because the bar and the body move independently in the PC but move as a single unit in the JS and S. Consequently, the mass of the body was included in the total system mass for both the JS and the S and, thus, changed the external load (as a percentage of 1RM) applied for each intensity. Adequate rest (at least 4 d) was provided between the each of the four testing sessions.

1RM testing. Dynamic strength testing during the first session was performed in both the S and PC in a randomized order with a 30-min break between the two tests. A 1RM was estimated for each subject on the basis of body weight for each lift. The subject then performed a series of warm-up sets and several maximal lift attempts until a 1RM for each respective lift was obtained. This protocol has been used extensively in our laboratory (28).

Jump squat testing. Participants set up for the JS in a standing position while holding a barbell across their shoulders. After instruction, subjects initiated the JS via a downward countermovement to a visually monitored knee angle of approximately 90° . Participants were instructed to keep constant downward pressure on the barbell throughout the jump and were encouraged to reach a maximum jump height with every trial in an attempt to maximize power output (2,21). The bar was not to leave the shoulders of the subject. If these requirements were not met, the trial was repeated. Because body mass must be moved in addition to the external load during a JS, the resulting force, velocity, and power were determined by the athlete's ability to accelerate the total system mass (i.e., external load + body mass). However, the lower leg and feet remain relatively static during the phase of the lift at which peak power typically occurs (specifically, just before take-off). As a result, the mass of the shanks (12% of body mass (22)) were excluded from the system mass used in determining the loading conditions and subsequent power calculations (6). Therefore, loading conditions were determined as the sum of a percentage of 1RM plus the

subject's body mass minus their shank mass (% of MDS = %1RM + [body mass - shank mass]) (6,17). Subjects performed trials in a randomized fashion with 30, 40, 50, 60, 70, 80, and 90% of MDS. These loading conditions correspond to 0, 12, 27, 42, 56, 71, and 85% of 1RM (i.e., percentage of external load only). Adequate recovery was provided between all trials (2 min for 0 and 12% of 1RM and 3 min for 27, 42, 56, 71, and 85% of 1RM).

Squat testing. During this session, subjects first performed an isometric squat test at a knee angle of 140° modified from Blazeovich et al. (3). Because peak force is typically generated later in the concentric phase of the squat (130–170° knee angle), this test was used to assess maximal force output at zero velocity (15). Participants then set up for the S in a standing position while holding a barbell across their shoulders. After instruction, the subjects initiated the squat via a downward countermovement to a visually monitored knee angle of approximately 90°. The concentric phase was performed by instructing the subject to move the bar as quickly as possible back to a standing position in an attempt to maximize power output (2,21). The subjects were instructed not to try to jump with the weight. The subject's feet did not leave the ground and the bar was not to leave the shoulders of the subject. If these requirements were not met, the trial was repeated. Subjects performed trials in a randomized fashion using the same loading and rest conditions as the JS.

Power clean testing. During this session, subjects first performed an isometric midhigh test at a knee angle of 140°, modified from Haff et al. (9). This test was used to assess maximal isometric force output at zero velocity. Participants then set up for the PC in a standardized starting position with a shoulder-width grip. Bar path was monitored for each trial to ensure proper form as described by Stone et al. (24). Subjects were instructed to use maximal effort for each lift, regardless of the load, to ensure maximal force application and maximal bar velocity in the second pull (2,21). Subjects were instructed not to leave the ground or to attempt to jump with the weight. If these requirements were not satisfied, the trial was repeated. The PC differs from the JS and S in that the bar moves in-

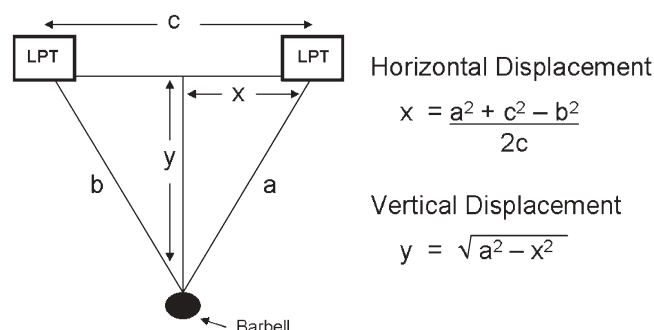


FIGURE 1—Two LPT configuration used for the calculation of vertical and horizontal displacement; a and b , voltage output from LPT converted to displacement values through calibration to known distances; c , constant distance between LPT; x , horizontal displacement; y , vertical displacement.

Jump Squat

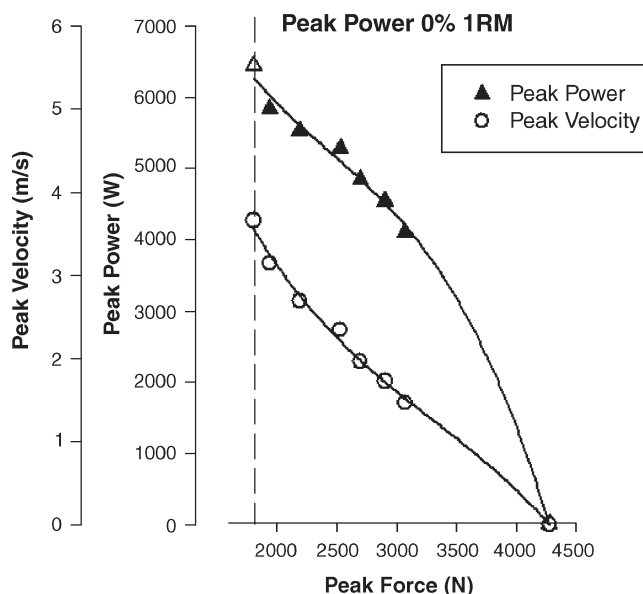


FIGURE 2—Force-velocity and force-power plots for the JS. Absolute peak power is designated by solid triangles; peak velocity is denoted by open circles. Optimal load is identified by an open triangle.

dependently of the body. Even though the PC involves moving the body from a squatting to a standing position, this movement does not directly influence the acceleration of the bar and, thus, does not affect the resulting power output of the lift (7). As a result, the loading parameters were not determined in the same manner as the JS and S. Subjects performed trials in a randomized fashion with 30, 40, 50, 60, 70, 80, and 90% of their 1RM in the PC (13,14). Adequate recovery was provided between all trials (2 min for 30 and 40% of 1RM, and 3 min for 50, 60, 70, 80, and 90% of 1RM).

Analog data collection and analysis. All testing (1RM, isometric squat, isometric midhigh pull, JS, S, and PC) was performed with the subjects standing on a force plate (AMTI, BP6001200, Watertown, MA) with the barbell (during the 1RM, JS, S and PC) attached to two linear position transducers (LPT) (Celesco Transducer Products, PT5A-150, Chatsworth, CA) (Fig. 1). The combined retraction tension of the two LPT was 16.3 N; this was accounted for in all subsequent calculations. The three analog signals were collected for every trial during each testing session at 1000 Hz using a BNC-2010 interface box with an analog-to-digital card (National Instruments, NI PCI-6014, Austin, TX). Customized software (LabVIEW, National Instruments, Version 7.1, Austin, TX) was used for recording and analyzing the data. Signals from the two LPT and the force plate underwent rectangular smoothing with a moving average half-width of 12. From laboratory calibrations, the two LPT's voltage outputs were converted into displacement in meters, and the force plate voltage output was converted into vertical force in newtons. The two LPT were mounted above and anterior, and above and posterior with respect to the

Squat

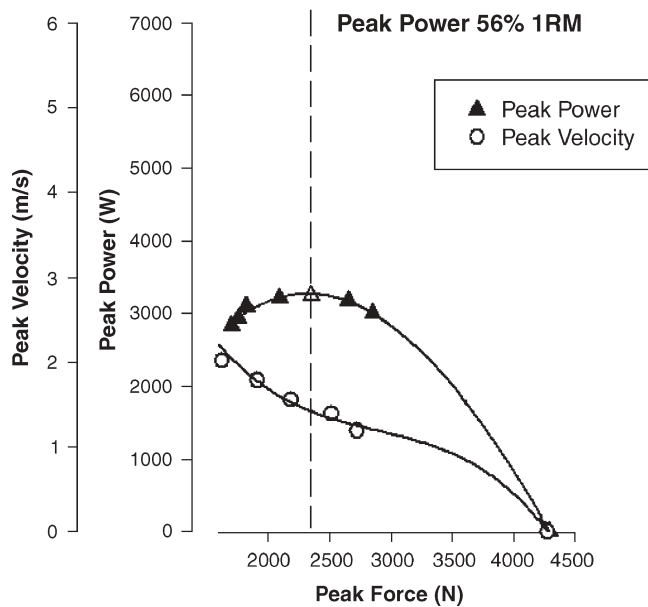


FIGURE 3—Force-velocity and force-power plots for the S. Absolute peak power is designated by *solid triangles*; peak velocity is denoted by *open circles*. Optimal load is identified by an *open triangle*.

subject, forming a triangle when attached to the barbell. Combining the known distances between the 2 LPT, in conjunction with the displacement measurements of both LPT as a result of barbell movement, allowed for the calculation of vertical and horizontal displacement of the barbell. Bar velocity was derived from the displacement and time data, and ground reaction force was measured directly by the force plate. Power was calculated from the

Power Clean

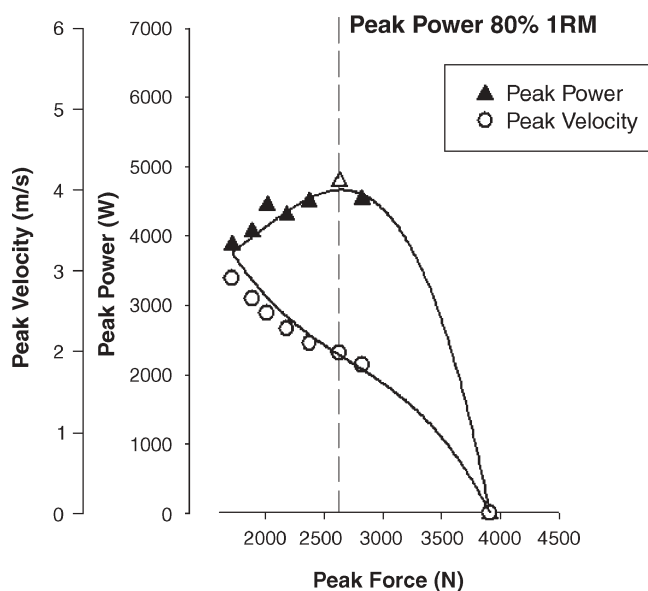
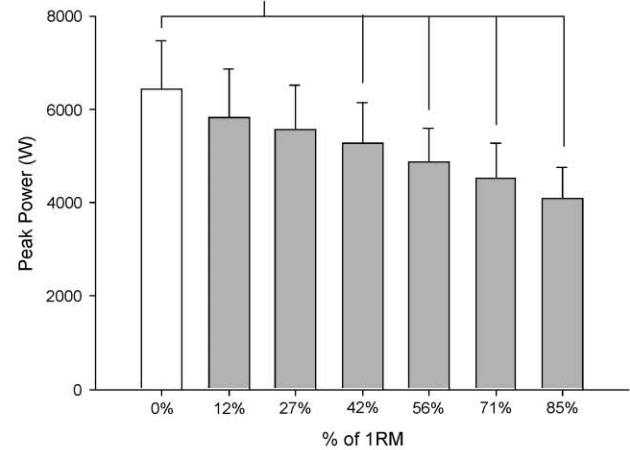


FIGURE 4—Force-velocity and force-power plots for the PC. Absolute peak power is designated by *solid triangles*; peak velocity is denoted by *open circles*. Optimal load is identified by an *open triangle*.

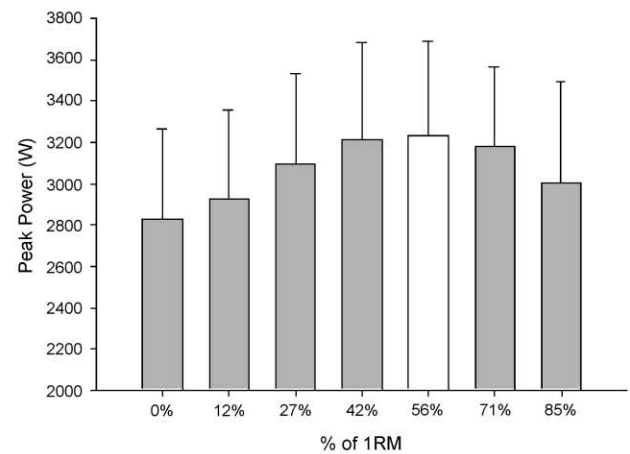
product of the velocity and force data (intraclass correlation coefficient (ICC) of peak power at the optimal load in the JS = 0.95, S = 0.94, and PC = 0.98; all are significant at $P < 0.01$).

Statistical analysis. A general linear model with repeated measures and Bonferoni *post hoc* tests were used

Jump Squat



Squat



Power Clean

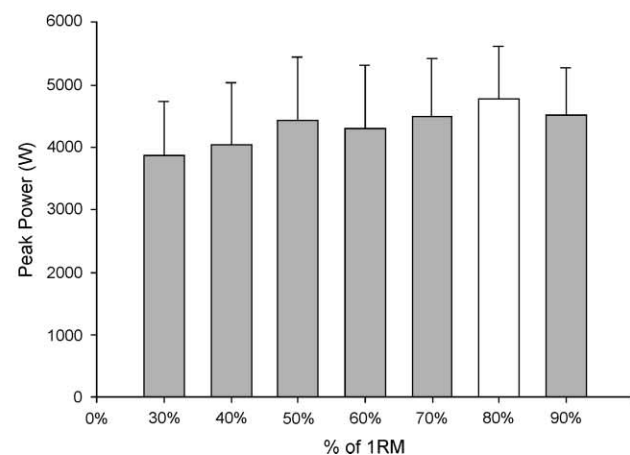


FIGURE 5—Absolute peak power output across the loading spectrum in the JS, S, and PC. * Significant ($P \leq 0.05$) difference between peak power at the optimal load (designated by *open bar*) and absolute peak power at other loads within the same lift.

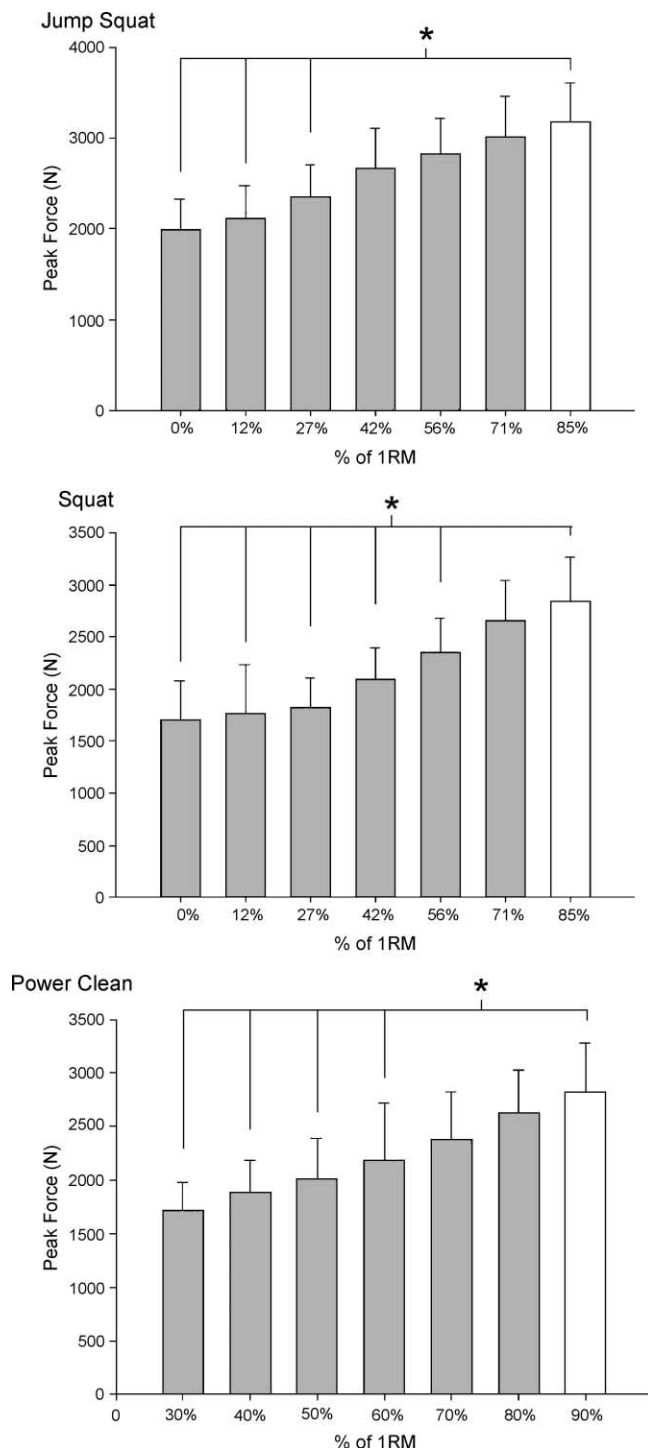


FIGURE 6—Peak force across the loading spectrum in the JS, S, and PC. * Significant ($P \leq 0.05$) difference between load that maximized peak force (designated by open bar) and the peak force at other loads within the same lift.

to determine whether there were significant differences within and between conditions for velocity, force, and power. Statistical significance for all analyses was defined by $P \leq 0.05$. Results are summarized as means $\pm$ standard deviations. Effect sizes were estimated at $\eta^2 = 0.435$, 0.546, and 0.943 at observed power levels of 1.000, 1.000, and 1.000 for peak power (PP), peak force (PF), and peak velocity (PV), respectively, for the JS. Effect sizes were

estimated at $\eta^2 = 0.095$, 0.479, and 0.790 at observed power levels of 0.495, 1.000, and 1.000 for PP, PF, and PV, respectively, for the S. Effect sizes were estimated at $\eta^2 = 0.101$, 0.579, and 0.879 at observed power levels of 0.528, 1.000, and 1.000 for PP, PF, and PV, respectively, for the PC. All statistical analyses were performed through

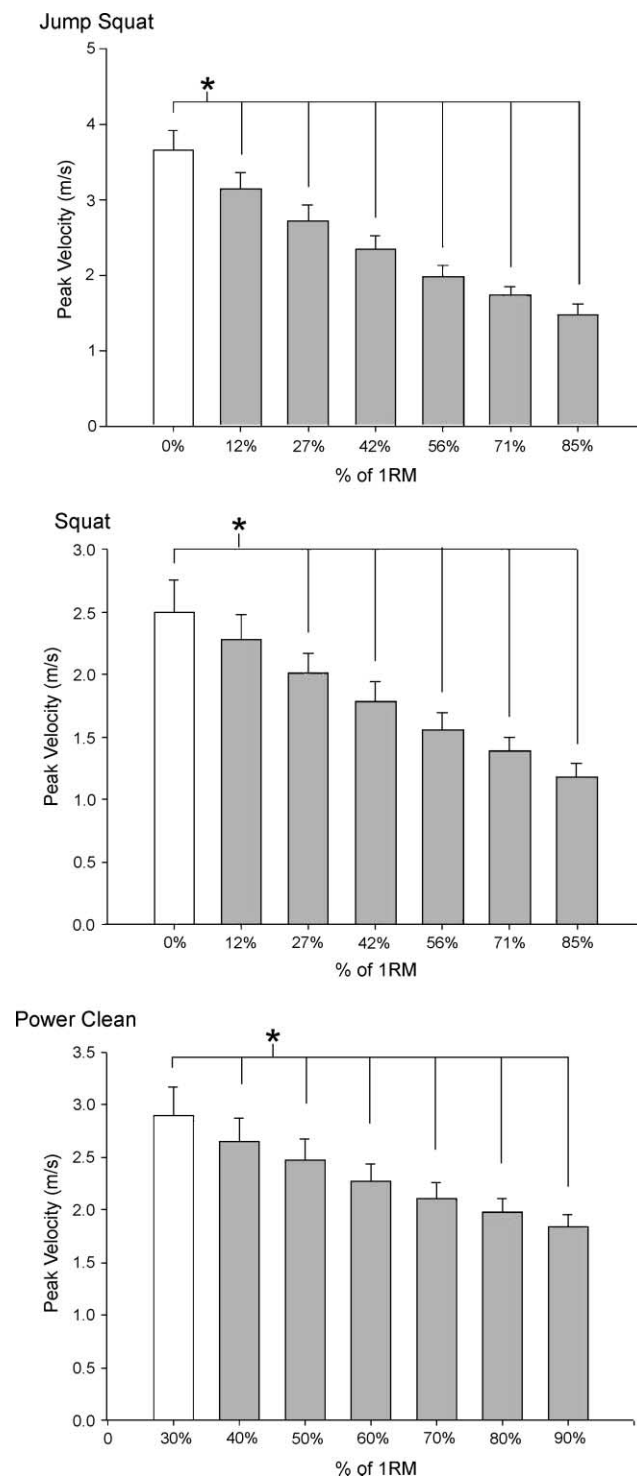


FIGURE 7—Peak velocity across the loading spectrum in the JS, S, and PC. * Significant ($P \leq 0.05$) difference between load that maximized peak velocity (designated by open bar) and the peak force at other loads within the same lift.

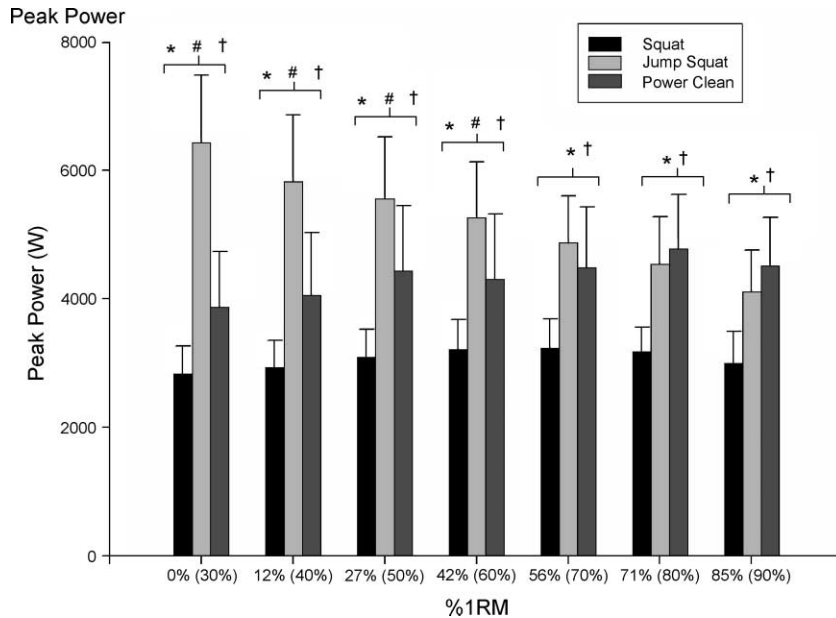


FIGURE 8—Comparison of absolute peak power output in the JS (light bars), S (black bars), and the PC (gray bars) across the loading spectrum. Loads expressed as % 1RM in the JS and S (% 1RM in the PC). * Significant ($P \leq 0.05$) difference between JS and S; # significant ($P \leq 0.05$) difference between JS and PC; † significant ($P \leq 0.05$) difference between S and PC.

the use of a statistical software package (SPSS, Version 11.0, SPSS Inc., Chicago, IL).

RESULTS

Jump squat. The optimal load for the JS was 0% of 1RM (30% of MDS); although not statistically different from all other intensities, this load elicited a greater peak power output than any other load (Fig. 2). Absolute peak power was significantly lower at 27% of 1RM in addition to 42, 56, 71, and 85% of 1RM. Maximum force was obtained at the heaviest loading condition (85% of 1RM), which was significantly greater than 0, 12, and 27% of 1RM (Fig. 6). Peak velocity was maximized during the body weight JS (0% of 1RM) (Fig. 2). Peak velocity at 27, 42, 56, 71, and 85% of 1RM was significantly lower than 0% of 1RM (Fig. 7).

Squat. The load that maximized power output in the S was 56% of 1RM (70% of MDS) (Fig. 3). However, peak power was not significantly different across the loading spectrum (Fig. 5). Force was maximized at the heaviest load—85% of 1RM—which was significantly different from 0, 12, 27, 42, and 56% of 1RM (Fig. 6). The 0% of 1RM loading condition (30% of MDS) elicited the maximum velocity (Fig. 3). Peak velocity at 0% of 1RM was greater than 27, 42, 56, 71, and 85% of 1RM (Fig. 7).

Power clean. Both absolute and relative peak power were greatest at 80% 1RM (Fig. 4). Peak power relative to body weight at 80% 1RM was significantly different from the loading conditions at 30 and 40% of 1RM. Examination of the loads that maximized force and velocity revealed similar patterns as both the JS and S. Force was maximized at 90% of 1RM and was significantly greater than 30, 40,

50, and 60% (Fig. 6). Bar velocity was the greatest at 30% of 1RM, with this load being significantly different from all other loading conditions (40, 50, 60, 70, 80, and 90% of 1RM) (Fig. 7).

Comparison of lifts. Power output differed between the JS, S, and PC throughout the loading spectrum (Fig. 8). The JS elicited greater power output than the S at 0, 12, 27, 42, 56, 71, and 85% of 1RM and the PC at 0, 12, 27, and 42% of 1RM (30, 40, 50, and 60% of 1RM for the PC). Additionally, higher power output was observed in the PC compared with the S at 0, 12, 27, 42, 56, 71, and 85% of 1RM (30, 40, 50, 60, 70, 80, 90% of 1RM for the PC). Peak force during the JS was significantly higher than both the S and PC at 42 and 56% of 1RM (60 and 70% of 1RM for the PC) as well as at 27% of 1RM in the S (Fig. 9).

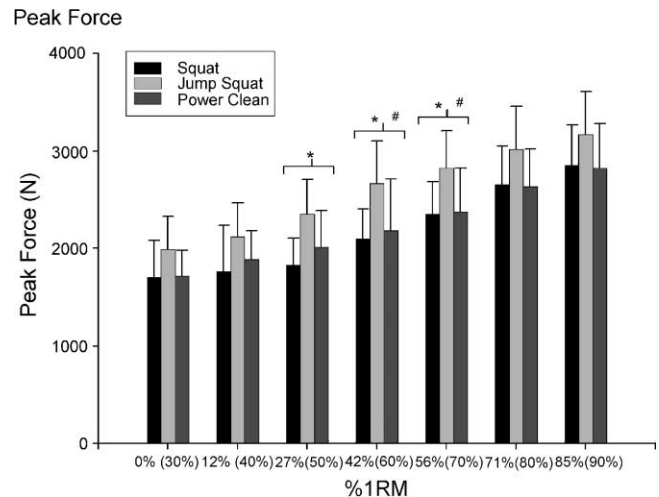


FIGURE 9—Comparison of peak force produced in the JS (light bars), S (black bars), and the PC (gray bars) across the loading spectrum. Loads expressed as % 1RM in the JS and S (% 1RM in the PC). * Significant ($P \leq 0.05$) difference between JS and S; # significant ($P \leq 0.05$) difference between JS and PC.

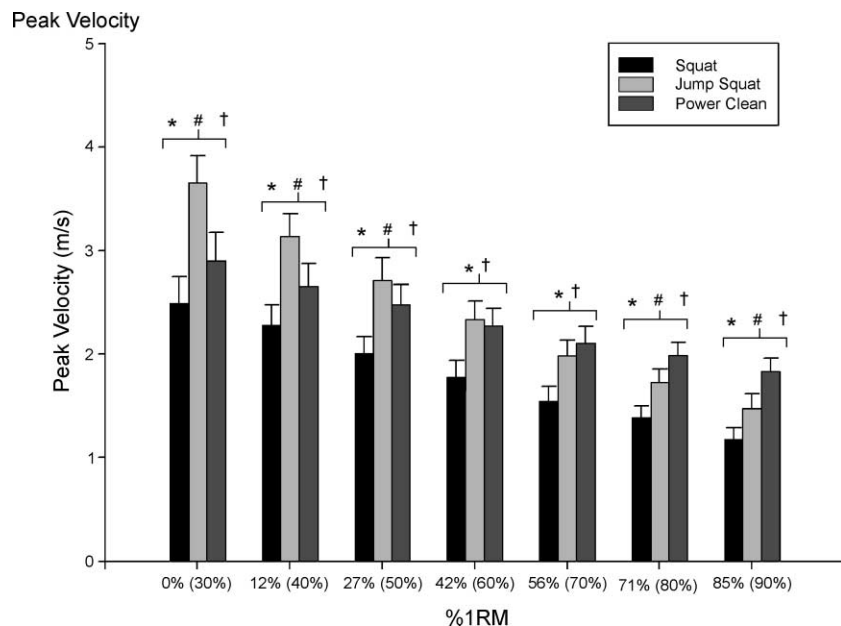


FIGURE 10—Comparison of peak velocity produced in the JS (light bars), S (black bars), and the PC (gray bars) across the loading spectrum. Loads expressed as % 1RM in the JS and S (% 1RM in the PC). * Significant ($P \leq 0.05$) difference between JS and S; # significant ($P \leq 0.05$) difference between JS and PC; † significant ($P \leq 0.05$) difference between S and PC.

Both the JS and the PC displayed greater velocities than the S across all loading conditions (Fig. 10). In comparing peak velocity between the JS and PC, Figure 9 displays that greater velocities were achieved in the JS at the lighter loads: 0, 12, and 27% of 1RM (30, 40, and 50% of 1RM for the PC). However, during the heavy loading conditions—71 and 85% of 1RM (80 and 90% of 1RM for the PC)—a shift occurred in which the PC produced greater peak velocities than the JS ($P \leq 0.05$).

DISCUSSION

As hypothesized, the results of the current investigation demonstrate the optimal load in the JS (0% 1RM; 30% of MDS), S (56% of 1RM; 70% of MDS), and PC (80% of 1RM; 80% of MDS) vary because of the nature of the movement involved. Furthermore, differences in the load–power relationship between the three lifts are established.

Jump squat. Considerable research has investigated power output during the JS (1,5,16,24). The current study's examination of the load–power relationship in the JS is novel in terms of the manner in which the loading parameters were established. The JS involves an athlete moving their body mass in addition to the external load. Consequently, superior power output results from an enhanced ability to accelerate the total system mass (external load, plus body minus shank mass) (6). Thus, the external load was calculated as a percentage of the individual's MDS, which is equivalent to a percentage of their 1RM, plus their body minus shank mass. This method typically results in the athlete's body mass (i.e., 0% of 1RM) equaling approximately 30% of MDS and 85% of 1RM corresponding to 90% of MDS.

The optimal load in the JS was identified as 0% of 1RM (30% of MDS); although not statistically different from all other intensities, this load elicited the greatest power output of the examined loads. The 0% of 1RM load was light

enough for athletes to generate very high velocities (peak velocity: $3.66 \pm 0.26 \text{ m}\cdot\text{s}^{-1}$), and the body mass provided enough resistance to produce a substantial force output (peak force: $1990.54 \pm 338.55 \text{ N}$). Therefore, this load permitted the most favorable combination of force and velocity values, which, in turn, yielded the maximal mechanical power output. Although absolute peak power at 0% of 1RM (30% of MDS) was not significantly different from loads at 12 and 27% of 1RM (40 and 50% of MDS), the trend illustrated in Figure 2 clearly indicates that the optimal load occurs at 0% of 1RM. This trend is supported by the finding that relative peak power at 0% of 1RM (30% of MDS) was significantly different from 27% of 1RM (50% of MDS).

These findings are consistent with previous investigations of the JS. McBride et al. (16) evaluated JS performance in elite sprinters, Olympic lifters, power lifters, and untrained individuals. Power output was maximized in body weight (BW) jumps (0% of 1RM). Paralleling the trend observed in the current study, the optimal load (BW) was not statistically significant from countermovement jumps with external loads of 20 or 40 kg (16). More recently, Bourque and Sleivert (5) also reported the optimal load for power athletes (six male volleyball and two male badminton players) as 0% of 1RM. In addition, peak power at this load ($6117 \pm 867 \text{ W}$) was very similar to that found in the current investigation ($6437.14 \pm 1046.34 \text{ W}$). In contrast, Stone et al. (25) reported the optimal load for the JS to be 10% of 1RM. However, it is vital to acknowledge that 10% of 1RM was the lightest load examined, and thus it cannot be ruled out that the optimal load may have actually occurred at 0% of 1RM.

This result is also similar to investigations of the bench-press throw, another ballistic exercise. Newton and colleagues (20) examined power output in the bench-press throw across loads of 15, 30, 45, 60, 75, 90, and 100% MDS (which is synonymous with percentage of 1RM

because the system mass is equal to the external load only in upper-body movements). The highest power output was reported as being produced at 30% of MDS (corresponding to 0% of 1RM in the JS in the current study). Additionally, early investigations of single-joint movements such as elbow flexor movements have demonstrated that maximal power output occurred at loads of 30% of maximal isometric strength (12). Furthermore, this optimal load was found to be most effective in improving maximal mechanical power output, a finding that has been supported by more recent research (17,27).

Squat. The optimal load for the S occurred at 56% of 1RM (70% of MDS). Peak power at the optimal load was very similar to loads of 42 and 71% of 1RM (Fig. 5). These findings are akin to the literature examining power output in the S. Most recently Izquierdo and colleagues (11) defined the optimal load in a concentric-only S to occur at 60% of 1RM for handball players, middle-distance runners, and untrained men, and at 45% of 1RM for weightlifters and road cyclists. The average power output of all subjects was very similar at both loads (861.0 and 860.5 W, respectively) as well as at 70% of 1RM (797.5 W) (11). Similarly, Siegel et al. (23) reported the optimal load in the S to occur at approximately 60% of 1RM, with high power outputs continuing from 50 to 80% of 1RM. Studies investigating the S and concentric-only S in middle-aged and elderly men also have reported the optimal load to occur at 60 and 70% of 1RM, respectively (10,11).

Each of these investigations provided similar instructions as the present study by asking subjects to complete the lift in explosive manner. Whereas the S is a nonballistic movement that involves a deceleration phase, when performed in an explosive manner, it becomes more analogous to ballistic movements such as the JS. It is hypothesized that if athletes were instructed to perform the movement less explosively, the optimal load would occur at a heavier load (i.e., possibly 80 or 90% of 1RM). The diminished difference in velocity between moderate and heavy loads, coupled with force output increasing relative to the intensity, would account for this shift toward higher percentages of 1RM. Therefore, the theory that the deceleration phase of the S would exceed that of both the PC and JS, resulting in an optimal load at a heavier intensity, may be evident if the movement were performed at a self-selected pace. The influence of the deceleration phase on the load–power relationship has not been directly investigated in the S (i.e., comparison of power output in the S performed explosively and slowly). However, previous study comparing ballistic and nonballistic movements have revealed results paralleling these theorized outcomes (21).

Power clean. The present study identified the optimal load in the PC to occur at 80% of 1RM. Peak power at this load (4786.63 ± 835.91 W) was within 485.6 W of the peak power across loads of 50 to 90% of 1RM. These results are similar to two recent investigations examining the influence of load on power output in the PC (14,28). Kawamori and associates (14) established that peak power in the hang clean was optimized at 70% of 1RM but was not

significantly different from peak power at 50, 60, 80, and 90% of 1RM. Relative peak power across these loads was within $4.53 \text{ W} \cdot \text{kg}^{-1}$, which is similar to the range of the current study ($5.74 \text{ W} \cdot \text{kg}^{-1}$). It is important to note that in Kawamori et al. (14), power was measured through the use of a force plate only. Because the PC involves the bar moving independently of the body, the system velocity measured by the force plate may not be representative of the actual bar velocity. Therefore, it is possible that the force plate methodology misrepresents power output during the PC. This is supported by data from this laboratory that highlight differences in the velocity and power values measured from a force plate and the two-potentiometer and force plate system used in the present study.

Winchester et al. (28) compared power output in the PC at 50, 70, and 90% 1RM and reported maximal power outputs to occur at 70% of 1RM before 4 wk of PC training and 50% of 1RM after training. No significant differences were reported between the loads, because peak power output across the three loads only differed by 466.83 and 520.35 W before and after training, respectively. Additionally, equivalent to the current study, an investigation of the midhigh pull revealed the optimal load as 80% of 1RM (8). The midhigh pull is a modified version of the hang PC, without the catch that involves the segment of the PC, in which peak power typically occurs (i.e., during the second pull phase before the catch phase). However, this study is limited in that it only used loads equivalent to 80, 90, and 100% of 1RM, and significant differences between the loads were not attained. Therefore, the optimal load of 80% of 1RM and the similarity of power outputs across loads of 50–90% of 1RM in the PC observed in the current study are analogous to previous research. This suggests that the optimal load in the PC occurs at a submaximal load within this range.

Influence of methodology. Also of importance in this investigation is the methodology used to collect and analyze power output. Several studies have used only a single linear position transducer (LPT) attached to a barbell to calculate power through the double differentiation of displacement data in both the JS and S (1,5). It is hypothesized that such displacement-based systems may lead to a misrepresentation of the maximum power output during explosive athletic movements. This hypothesis is based on the fact that there are several concerns with this methodology, including the misrepresentation of force–time data through double differentiation (29), horizontal movement not being accounted for, and differences between force measured at the bar (representing external load only) and the actual force exerted by the system (6). These aspects are addressed in the data-collection system used in the current study, suggesting that the technique allows for a more accurate measure of power during multidimensional, dynamic movements. Furthermore, on the basis of data from this laboratory, differences exist in power output during a JS between the displacement-based system and methodologies involving both displacement and force measurements.

Comparison of lifts. The most unique aspect of this investigation is the comparison of power output between the three lifts. Although much research has been undertaken on the lifts individually, questions still remain regarding the effect of exercise type on power production. This is particularly true for the comparison of ballistic and Olympic lifts. Power output is influenced by the nature of the movement involved in the lift; thus, the optimal loads of the JS, S, and PC are different. In the traditional S, it is necessary to decelerate the bar towards the end of the range of motion (13). However, the JS is a ballistic exercise in which the load is actually released at the end of the range of motion, resulting in continued acceleration throughout the movement. High forces are generated in light-load situations with ballistic movements because of the high acceleration rates throughout the movement (21). Therefore, ballistic movements such as the JS result in maximal power output occurring at lighter loads than the traditional resistance training exercises such as the S. The results of this investigation support this theory, with the optimal load in the S occurring at 56% of 1RM as opposed to 0% of 1RM in the JS (Figs. 2 and 3). Furthermore, because of the high-force and high-velocity movement involved with Olympic lifts such as the PC, high power outputs may occur at heavier loads than both the S and JS (8). As expected, the optimal load in the PC (80% of 1RM) occurred closer to MDS than the optimal load in both the JS and the S (Fig. 8).

These comparisons provide some insight into the specificity of power training available to meet the on-field demands of various sports. The data indicate that both the JS and PC are far superior to the S in eliciting high power outputs across the loading spectrum (Fig. 8). Although more power is generated in the JS than the PC at lighter loads, there is no difference between the two exercises under heavy loading conditions. There is, however, a trend toward the PC

eliciting greater power production at 80 and 90% of 1RM. This indicates that the JS may be the most specific exercise mode for athletes required to produce high velocities against light loads, such as sprinters and jumpers. In contrast, the PC may be the most effective training exercise for athletes required to generate high velocities against heavy loads, for instance, weightlifters and football linemen.

CONCLUSIONS

Although much debate surrounds the acute variables in power-training programs (i.e., repetitions, sets, recovery intervals), there is a consensus within the literature regarding the most efficient training load. When designing a program to improve maximal muscular power, individuals should train using the load at which muscle power output is maximized—the optimal load—because training at this load is most effective in improving maximal power output (12,13,17–19, 27). The optimal load occurs at 0% of 1RM (30% of MDS) in the JS, 56% of 1RM (70% of MDS) in the S, and 80% of 1RM in the PC. Another vital aspect of designing the power phase of a periodized training program is the exercise used in power training. Both the JS and PC are far superior to the S in eliciting high power outputs regardless of the load applied. The JS should be used for athletes required to move light loads at high velocities, for instance, sprinters, jumpers, and basketball and volleyball players. In contrast, the PC is ideal for athletes whose on-field demands dictate the need to move heavy loads as quickly as possible, for instance, Olympic weightlifters and football linemen. Further study is required to examine the most effective repetition, set, and rest combination for training to improve maximal power output.

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REFERENCES

1. BAKER, D., S. NANCE, and M. MOORE. The load that maximizes the average mechanical power output during jump squats in power trained athletes. *J. Strength Cond. Res.* 15:92–97, 2001.
2. BEHM, D. G., and D. G. SALE. Intended rather than actual movement velocity determines velocity-specific training response. *J. Appl. Physiol.* 74:359–368, 1993.
3. BLAZEVOICH, A. J., N. GILL, and R. U. NEWTON. Reliability and validity of two isometric squat tests. *J. Strength Cond. Res.* 16: 298–304, 2002.
4. BOTTINELLI, R., M. A. PELLEGRINO, M. CANEPARI, R. ROSSI, and C. REGGIANI. Specific contributions of various muscle fibre types to human muscle performance: an in vitro study. *J. Electromyogr. Kinesiol.* 9:87–95, 1999.
5. BOURQUE, J. P. Determinants of load at peak power during maximal effort squat jumps in endurance and power trained athletes [dissertation]. Fredericton, Canada: Department of Kinesiology, University of New Brunswick, 2003.
6. DUGAN, E. L., T. L. A. DOYLE, B. HUMPHRIES, C. HASSON, and R. U. NEWTON. Determining the optimal load for jump squats: a review of methods and calculations. *J. Strength Cond. Res.* 18:668–674, 2004.
7. GARHAMMER, J. A review of power output studies of Olympic and power lifting: methodology, performance prediction, and evaluation tests. *J. Strength Cond. Res.* 7:76–89, 1993.
8. HAFF, G. G., M. H. STONE, H. S. O'BRYANT, et al. Force-time dependent characteristics of dynamic and isometric muscle actions. *J. Strength Cond. Res.* 11:269–272, 1997.
9. HAFF, G. G., J. M. CARLOCK, M. J. HARTMAN, et al. Force-time curve characteristics of dynamic and isometric muscle actions of elite women olympic weightlifters. *J. Strength Cond. Res.* 19: 741–748, 2005.
10. IZQUIERDO, M., J. IBANEZ, E. GOROSTIAGA, et al. Maximal strength and power characteristics in isometric and dynamic actions of the upper and lower extremities in middle-aged and older men. *Acta Physiol. Scand.* 167:57–68, 1999.
11. IZQUIERDO, M., K. HÄKKINEN, J. J. GONZALEZ-BADILLO, J. IBANEZ, and E. M. GOROSTIAGA. Effects of long-term training specificity on maximal strength and power of the upper and lower extremities in athletes from different sports. *Eur. J. Appl. Physiol.* 87:264–271, 2002.
12. KANEKO, M., T. FUCHIMOTO, H. TOJI, and K. SUEI. Training effect of different loads on the force-velocity relationship and mechanical power output in human muscle. *Scand. J. Med. Sci. Sports* 5:50–55, 1983.
13. KAWAMORI, N., and G. G. HAFF. The optimal training load for the development of muscular power. *J. Strength Cond. Res.* 18: 675–684, 2004.
14. KAWAMORI, N., A. J. CRUM, P. A. BLUMERT, et al. Influence of

- different relative intensities on power output during the hang power clean: identification of the optimal load. *J. Strength Cond. Res.* 9:698–708, 2005.
15. KELLIS, E., F. ARAMBATZI, and C. PAPADOPOULOS. Effects of load on ground reaction force and lower limb kinematics during concentric squats. *J. Sport Sci.* 23:1045–1055, 2005.
 16. McBRIDE, J. M., T. TRIPLETT-McBRIDE, A. DAVIE, and R. U. NEWTON. A comparison of strength and power characteristics between power lifters, Olympic lifters and sprinters. *J. Strength Cond. Res.* 13:58–66, 1999.
 17. BRIDE, J. M., T. TRIPLETT-McBRIDE, A. DAVIE, and R. U. NEWTON. The effect of heavy- vs. light-load jump squats on the development of strength, power, and speed. *J. Strength Cond. Res.* 16: 75–82, 2002.
 18. MORITANI, T., M. MURO, K. ISHIDA, and S. TAGUCHI. Electrophysiological analyses of the effects of muscle power training. *Res. J. Phys. Educ. Japan* 1:23–32, 1987.
 19. MORITANI, T. Neuromuscular adaptations during the acquisition of muscle strength, power and motor skills. *J. Biomech.* 26:95–107, 1993.
 20. NEWTON, R. U., A. MURPHY, B. J. HUMPHRIES, G. J. WILSON, W. J. KRAEMER, and K. HÄKKINEN. Influence of load and stretch shortening cycle on the kinematics, kinetics and muscle activation that occurs during explosive upper-body movements. *Eur. J. Appl. Physiol. Occup. Physiol.* 75:333–342, 1997.
 21. NEWTON, R. U., W. J. KRAEMER, K. HÄKKINEN, B. J. HUMPHRIES, and A. J. MURPHY. Kinematics, kinetics and muscle activation during explosive upper body movements. *J. Appl. Biomech.* 12:31–43, 1996.
 22. PISCOPO, J., and J. BAILEY. *Kinesiology: The Science of Movement*. New York, NY: John Wiley and Sons, 1981, p. 128.
 23. SIEGEL, J. A., R. M. GILDERS, R. S. STARON, and F. C. HAGERMAN. Human muscle power output during upper- and lower-body exercises. *J. Strength Cond. Res.* 16:173–178, 2002.
 24. STONE, M. H., H. S. O'BRYANT, F. E. WILLIAMS, R. L. JOHNSON, and K. C. PIERCE. Analysis of bar paths during the snatch in elite male weightlifters. *Strength Cond. J.* 20:30–38, 1998.
 25. STONE, M. H., H. S. O'BRYANT, L. MCCOY, R. COGLIANESE, M. LEHMKUHL, and B. SCHILLING. Power and maximum strength relationships during performance of dynamic and static weighted jumps. *J. Strength Cond. Res.* 17:140–147, 2003.
 26. VAN LEEUWEN, J. L. Optimum power output and structural design of sarcomeres. *J. Theor. Biol.* 149:229–256, 1991.
 27. WILSON, G. J., R. U. NEWTON, A. J. MURPHY, and B. J. HUMPHRIES. The optimal training load for the development of dynamic athletic performance. *Med. Sci. Sports Exerc.* 25:1279–1286, 1993.
 28. WINCHESTER, J. B., T. M. ERICKSON, J. B. BLAAK, and J. M. McBRIDE. Changes in bar-path kinematics and kinetics after power-clean training. *J. Strength Cond. Res.* 19:177–183, 2005.
 29. WOOD, G. A. Data smoothing and differentiation procedures in biomechanics. *Exerc. Sport Sci. Rev.* 10:308–362, 1982.